9.2 Character set

Character set is a set of characters which are used to write a C++ program. A character represents any alphabet, number or any other symbol (special characters) mostly available in the keyboard. C++ accepts the following characters.

Alphabets	A …. Z, a ….. z
Numeric	0 …. 9
Special Characters	+ - * / ~ ! @ # $ % ^& [ ] ( ) { } = >< _ \ | ? . , : ‘ “ ;
White space	Blank space, Horizontal tab (→), Carriage return (↲), Newline, Form feed
Other characters	C++ can process any of the 256 ASCII characters as data.

9.3 Lexical Units (Tokens):

C++ program statements are constructed by many different small elements such as commands, variables, constants and many more symbols called as operators and punctuators. These individual elements are collectively called as **Lexical units** or **Lexical elements** or **Tokens**. C++ has the following tokens:

- Keywords
- Literals
- Punctuators
- Identifiers
- Operators

TOKEN:

The smallest individual unit in a program is known as a Token or a Lexical unit

9.3.1 Keywords

Keywords are the reserved words which convey specific meaning to the C++ compiler. They are the essential elements to construct C++ programs. Most of the keywords are common to C, C++ and Java.

C++ is a case sensitive programming language so, all the keywords must be in lowercase.

Table 9.1 C++ Keywords

asm	auto	break	case	catch
char	class	const	continue	default
delete	do	double	else	enum
extern	float	for	friend	goto
if	inline	int	long	new
operator	private	protected	public	register
return	short	signed	sizeof	static
struct	switch	template	this	throw
try	typedef	union	unsigned	virtual
void	volatile	while		

- With revisions and additions, the recent list of keywords also includes:

using, namespace, bal, static_cast, const_cast, dynamic_cast, true, false

- Identifiers containing a double underscore are reserved for use by C++ implementations and standard libraries and should be avoided by users.

9.3.2 Identifiers

Identifiers are the user-defined names given to different parts of the C++ program viz. variables, functions, arrays, classes etc., These are the fundamental building blocks of a program. Every language has specific rules for naming the identifiers.

Rules for naming an identifier:

- The first character of an identifier must be an alphabet or an underscore (_).
- Only alphabets, digits and underscore are permitted. Other special characters are not allowed as part of an identifier.
- C++ is case sensitive as it treats upper and lower-case characters differently.
- Reserved words or keywords cannot be used as an identifier name.

11th Std Computer Science_EM Chapter 9.indd 116 06-01-2023 15:08:44